

Question number	Answer	Additional guidance	Mark
7(a)	A description that includes three of the following points: • by carbon fixation (1) • resulting in the formation of GP (1) • resulting in the formation of GALP / glucose / hexose (1) • which is used to make (polymer) organic molecules (1)	ACCEPT description e.g. carbon dioxide {binds to / reacts with} RuBP ACCEPT other alternative names for GP ACCEPT other alternative names for GALP and glucose ACCEPT named organic molecule e.g. cellulose	(3)

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7(b)(i)	An explanation that includes the following points: • increase in temperature speeds up RUBISCO / more carbon dioxide enters (leaf) <u>by diffusion</u> (1) • so {carbon fixation / light-independent reactions / Calvin cycle) faster (1)	ACCEPT converse / in context of denaturing RUBISCO and carbon fixation stopping ACCEPT description e.g. GP produced faster	(2)

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7(b)(ii)	An explanation that includes two of the following points: - increased rate of respiration (in plant roots) (1) - faster {active transport / uptake} of (mineral) ions (needed for photosynthesis) (1) - credit use of named ion in photosynthesis (1) OR - osmosis faster (into roots) (1) - therefore more water transported to leaves in xylem (1) - therefore more water for {photolysis / light-dependent reactions} (1) OR - increased membrane permeability (of plant roots) (1) - increased {osmosis / diffusion of (mineral) ions (1) - therefore more water for {photolysis / light-dependent reactions / credit use of named ion in photosynthesis (1)	ACCEPT converse for lower temperature IGNORE metabolism DO NOT ACCEPT photorespiration ACCEPT named mineral ion needed for photosynthesis e.g. magnesium ions, nitrates, sulfates, phosphates IGNORE diffusion, nutrients, easier ACCEPT more decomposition providing more mineral in soil	(2)

Question number	Answer	Additional guidance	Mark
7(c)	An explanation that includes the following points: • less water for photolysis (1) • less ATP and {reduced NADP / NADPH} (1) • limiting the rate of the {light-independent reactions / Calvin cycle} (1)	NB must be in context of what would happen with less water for 3 marks to be awarded	(3)

Question number	Answer	Additional guidance	Mark
7(d)	An explanation that includes four of the following points: • AM increase {biomass / nitrate availability} in high nitrate soil and decrease {biomass / nitrate availability} in low nitrate soil (1) • (decrease) due to {cell death / decomposition / respiration greater than photosynthesis} (1) • ECM is able to {provide more nitrate ions / make more nitrate available} (to plants) (1) • nitrates are used (by plants) in production of {amino acids / protein} (1) • {nitrates / named nitrogen-containing molecule} needed for plant {biomass production / growth} (1)	NB piece together ACCEPT description of how nitrates are made more available IGNORE ECM produces more nitrates ACCEPT DNA / enzymes / plant growth factors	(4)